

# ArithmeticSite

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June 1, 2026

# Chapter 1

## The multiplicative monoid $\mathbb{N}^\times$

The Arithmetic Site, introduced by Connes and Consani, is a geometric object — a topos equipped with a structure sheaf — whose points in the topos-theoretic sense are in natural bijection with the adèle class space of  $\mathbb{Q}$ . Its construction begins with a single, elementary monoid: the positive integers under multiplication. Everything else — the tropical semiring, the presheaf topos, the structure sheaf, and ultimately the connection to the Riemann zeta function — is built on top of this one piece of data.

**Definition 1.1.** Let  $\mathbb{N}^\times$  denote the set  $\{1, 2, 3, \dots\}$  of strictly positive integers equipped with ordinary multiplication. This is a commutative monoid with unit 1. In Lean we represent this as `PNat` with its multiplicative structure, via the abbreviation `NPos`.

The unique factorisation theorem says that  $\mathbb{N}^\times$  is the free commutative monoid on the set of primes: every element factors uniquely as a finite product  $\prod_p p^{a_p}$ . This freeness will not be used explicitly in what follows, but it is the arithmetic reason why the Arithmetic Site carries information about the prime numbers.

**Definition 1.2.** The *classifying category*  $B\mathbb{N}^\times$  of  $\mathbb{N}^\times$  is the one-object category whose single object is  $*$  and whose morphisms  $*$   $\rightarrow$   $*$  are the elements of  $\mathbb{N}^\times$ , with composition given by multiplication.

**Remark 1.3.** The passage from a monoid  $M$  to the one-object category  $BM$  is the standard *delooping* construction. A presheaf on  $BM$  — that is, a functor  $BM^{\text{op}} \rightarrow \mathbf{Set}$  — is exactly a set  $X$  equipped with a right action of  $M$ : each element  $n \in M$  acts as a function  $X \rightarrow X$ , and the action is compatible with composition. The presheaf topos  $\widehat{B\mathbb{N}^\times}$  introduced in Chapter 3 is precisely the category of such sets-with- $\mathbb{N}^\times$ -action, and it is in this topos that the Arithmetic Site lives.

## Chapter 2

# The tropical semiring $\bar{\mathbb{N}}$

**Definition 2.1.** The *tropical semiring*  $\bar{\mathbb{N}}$  is the set  $\mathbb{N} \cup \{+\infty\}$  equipped with:

- *Tropical addition:*  $a \oplus b := \min(a, b)$ , with identity element  $+\infty$ .
- *Tropical multiplication:*  $a \odot b := a + b$  (ordinary addition), with identity element 0.

In Lean,  $\bar{\mathbb{N}}$  is represented as `Tropical (WithTop ℕ)`, via the abbreviation `NBar`. The zero element is `trop ⊤` (i.e.  $+\infty$ ) and the multiplicative identity is `trop 0`.

**Lemma 2.2.**  $\bar{\mathbb{N}}$  is a commutative semiring.

*Proof.* This is the Mathlib instance `CommSemiring (Tropical (WithTop ℕ))`, found by `inferInstance`. □

**Lemma 2.3.** Tropical addition on  $\bar{\mathbb{N}}$  is idempotent:  $a \oplus a = a$  for all  $a \in \bar{\mathbb{N}}$ .

*Proof.*  $a \oplus a = \min(a, a) = a$ . In Lean this follows from `Tropical.add_eq_left`. □

**Definition 2.4.** For each  $n \in \mathbb{N}^\times$ , the *Frobenius endomorphism*  $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$  is defined by

$$\varphi_n(x) := n + x \quad (x \in \mathbb{N}), \quad \varphi_n(+\infty) := +\infty,$$

where  $+$  denotes ordinary addition in  $\mathbb{N} \cup \{+\infty\}$ . Equivalently, in `Tropical (WithTop ℕ)`, this is tropical multiplication by `trop (n : WithTop ℕ)`, i.e.  $\varphi_n(x) = \text{trop } n \odot x$ .

In Lean,  $\varphi_n$  is defined as an `AddMonoid.End NBar` via `frobeniusEndomorphism`.

**Lemma 2.5.** For each  $n \in \mathbb{N}^\times$ , the map  $\varphi_n : \bar{\mathbb{N}} \rightarrow \bar{\mathbb{N}}$  is an additive monoid endomorphism, i.e. it preserves tropical addition and the zero element  $+\infty$ :

$$\begin{aligned} \varphi_n(+\infty) &= +\infty, \\ \varphi_n(a \oplus b) &= \varphi_n(a) \oplus \varphi_n(b). \end{aligned}$$

The second identity follows from the distributivity  $n + \min(a, b) = \min(n + a, n + b)$ .

**Remark 2.6.** Note that  $\varphi_n$  is *not* a semiring homomorphism in general, since it does not preserve the multiplicative identity:  $\varphi_n(0) = n \neq 0$  for  $n > 0$ . It is however a homomorphism of  $\bar{\mathbb{N}}$ -modules.

**Lemma 2.7.** *The Frobenius endomorphisms satisfy the composition law*

$$\varphi_m \circ \varphi_n = \varphi_{m+n}$$

for all  $m, n \in \mathbb{N}^\times$ . In particular, the map  $n \mapsto \varphi_n$  is a semigroup homomorphism from  $(\mathbb{N}^\times, +)$  into the endomorphism semigroup  $\text{End}_{\text{AddMon}}(\bar{\mathbb{N}})$ .

*Proof.* For any  $x \in \bar{\mathbb{N}}$ :  $(\varphi_m \circ \varphi_n)(x) = m + (n + x) = (m + n) + x = \varphi_{m+n}(x)$ . In Lean this is `frobenius_comp`, proved via `mul_assoc` in `Tropical (WithTop ℕ)`.  $\square$

## Chapter 3

# The presheaf topos $\widehat{\mathbb{N}^\times}$

**Definition 3.1.** The *presheaf topos*  $\widehat{\mathbb{N}^\times}$  is the category of functors  $B\mathbb{N}^{\times\text{op}} \rightarrow \mathbf{Set}$ , i.e. the category of sets equipped with a left  $\mathbb{N}^\times$ -action. In Lean this is `Action (Type*) (Multiplicative ℕ+)`.

**Lemma 3.2.**  $\widehat{\mathbb{N}^\times}$  is a Grothendieck topos.

*Proof.* Any presheaf category on a small category is a Grothendieck topos. □

## Chapter 4

# The structure sheaf

**Definition 4.1.** The *structure sheaf*  $\mathcal{O}$  of the Arithmetic Site is the object of  $\widehat{\mathbb{N}^\times}$  whose underlying set is  $\bar{\mathbb{N}}$  and whose  $\mathbb{N}^\times$ -action is given by the Frobenius endomorphisms:  $n \cdot x := \varphi_n(x)$ .

**Lemma 4.2.** The assignment  $n \mapsto \varphi_n$  defines a semigroup homomorphism  $(\mathbb{N}^\times, +) \rightarrow \text{End}_{\text{AddMon}}(\bar{\mathbb{N}})$ , making  $\bar{\mathbb{N}}$  into a left  $(\mathbb{N}^\times, +)$ -module.

## Chapter 5

# The Arithmetic Site

**Definition 5.1.** The *Arithmetic Site* is the pair  $(\widehat{\mathbb{N}^\times}, \mathcal{O})$  consisting of the presheaf topos  $\widehat{\mathbb{N}^\times}$  together with its structure sheaf  $\mathcal{O}$ .

## Chapter 6

# Points of the topos

**Definition 6.1.** A *point* of a topos  $\mathcal{E}$  is a geometric morphism  $p : \mathbf{Set} \rightarrow \mathcal{E}$ , i.e. an adjoint pair  $p^* \dashv p_*$  where  $p^*$  is left exact.

**Theorem 6.2.** *The points of  $\widehat{\mathbb{N}^\times}$  are in natural bijection with the adèle class space  $\mathbb{Q}_+^\times \backslash \mathbb{A}^f / \widehat{\mathbb{Z}}^*$ .*

*Proof.* (sorry)

□

## Chapter 7

# The Frobenius correspondences

**Definition 7.1.** For each  $\lambda \in \mathbb{Q}_+^\times$ , the *Frobenius correspondence*  $\Phi_\lambda$  is a correspondence on  $\widehat{\mathbb{N}^\times} \times \widehat{\mathbb{N}^\times}$  arising from the scaling action of  $\mathbb{Q}_+^\times$  on the adèle class space.

**Theorem 7.2.** *The Frobenius correspondences realize the scaling action of  $\mathbb{Q}_+^\times$  on the adèle class space at the level of the topos  $\widehat{\mathbb{N}^\times}$ .*

*Proof.* (sorry)

□

## Chapter 8

# Connection to the Riemann zeta function

**Theorem 8.1.** *The Hasse–Weil zeta function of the Arithmetic Site recovers the Riemann zeta function  $\zeta(s)$ .*

*Proof.* (sorry)

□